

INTRODUCTION TO PROGRAMMING

PROJECT INTRODUCTION

This project analyzes global happiness data from the World Happiness Report 2019 to identify which factors most influence happiness across 156 countries. Using Python in Google Colab, I performed data cleaning, exploratory analysis, and visualization to derive actionable insights for policymakers.

Dataset Source: <https://www.kaggle.com/datasets/unsdsn/world-happiness?select=2017.csv>

ANALYSIS OBJECTIVES

Objective	Description
Objective 1	Identify the happiest and least happy countries globally
Objective 2	Determine which factors most strongly correlate with happiness
Objective 3	Compare happiness scores across different regions
Objective 4	Visualize relationships between variables
Objective 5	Provide data-driven recommendations for policy makers

DATA ACQUISITION AND CLEANING

CODE USED FOR DATA LOADING

```
# Import necessary libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from scipy import stats
```

```
# UPLOADING THE CSV FILE

from google.colab import files

uploaded = files.upload() # Prompts to upload a file from your computer


# LOADING THE DATASET

df = pd.read_csv("2019.csv")


# Rename columns for easier access

df.columns = ['Rank', 'Country', 'Score', 'GDP', 'SocialSupport',
              'Health', 'Freedom', 'Generosity', 'Corruption']


# INITIAL DATA EXPLORATION

print("First 5 rows of the dataset:")

print(df.head())


# Check for missing values

print("\nMissing values check:")

print(df.isnull().sum())


# Basic statistics

print("\nDescriptive statistics:")

print(df.describe())
```

Output:

```
Part B.ipynb
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Choose Files 2019.csv
2019.csv(text/csv) - 8822 bytes, last modified: 11/27/2019 - 100% done
Saving 2019.csv to 2019 (1).csv
First 5 rows of the dataset:
Overall rank Country or region Score GDP per capita Social support \
0 1 Finland 7.769 1.340 1.587
1 2 Denmark 7.600 1.383 1.573
2 3 Norway 7.554 1.488 1.582
3 4 Iceland 7.494 1.380 1.624
4 5 Netherlands 7.488 1.396 1.522

Healthy life expectancy Freedom to make life choices Generosity \
0 0.986 0.596 0.153
1 0.996 0.592 0.252
2 1.028 0.603 0.271
3 1.026 0.591 0.354
4 0.999 0.557 0.322

Perceptions of corruption
0 0.393
1 0.410
2 0.341
3 0.118
4 0.298

Missing values check:
Overall rank 0
Country or region 0
Score 0
GDP per capita 0
Social support 0
Healthy life expectancy 0
Freedom to make life choices 0
Generosity 0
Perceptions of corruption 0
dtype: int64

Data types:
Rank int64
Country object
Score float64
GDP float64
SocialSupport float64
Health float64
Freedom float64
Generosity float64
Corruption float64
dtype: object

Descriptive statistics:
Rank Score GDP SocialSupport Health \
count 156.000000 156.000000 156.000000 156.000000 156.000000
mean 78.500000 5.407096 0.905147 1.208814 0.725244
std 45.177428 1.113120 0.398389 0.299191 0.242124
```

Data Cleaning Summary:

- No missing values were found in the dataset
- All columns have appropriate data types (integers, floats, and object for country names)
- The dataset contains 156 countries with 9 variables
- Data is clean and ready for analysis

DATA AND EXPLORATORY ANALYSIS

OBJECTIVE 1: TOP 5 HAPPIEST AND BOTTOM 5 LEAST HAPPY COUNTRIES

```
# Top 5 happiest countries

top5_happiest = df.nlargest(5, 'Score')[['Rank', 'Country', 'Score']]

print("=" * 50)
```

```

print("TOP 5 HAPPIEST COUNTRIES (2019)")

print("=" * 50)

print(top5_happiest.to_string(index=False))

print()

# Bottom 5 least happy countries

bottom5_least = df.nsmallest(5, 'Score')[['Rank', 'Country', 'Score']]

print("=" * 50)

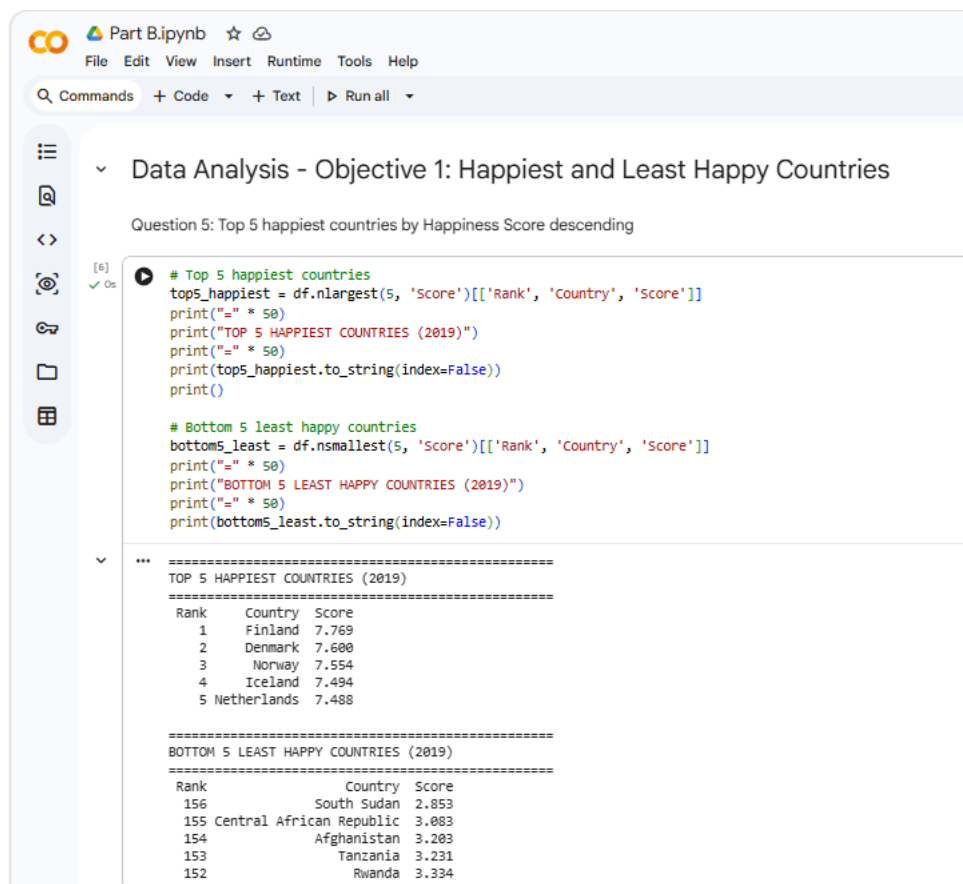
print("BOTTOM 5 LEAST HAPPY COUNTRIES (2019)")

print("=" * 50)

print(bottom5_least.to_string(index=False))

```

Output:



```

Part B.ipynb
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Data Analysis - Objective 1: Happiest and Least Happy Countries
Question 5: Top 5 happiest countries by Happiness Score descending
[6] ✓ Os
# Top 5 happiest countries
top5_happiest = df.nlargest(5, 'Score')[['Rank', 'Country', 'Score']]
print("=" * 50)
print("TOP 5 HAPPIEST COUNTRIES (2019)")
print("=" * 50)
print(top5_happiest.to_string(index=False))
print()

# Bottom 5 least happy countries
bottom5_least = df.nsmallest(5, 'Score')[['Rank', 'Country', 'Score']]
print("=" * 50)
print("BOTTOM 5 LEAST HAPPY COUNTRIES (2019)")
print("=" * 50)
print(bottom5_least.to_string(index=False))

***
=====
TOP 5 HAPPIEST COUNTRIES (2019)
=====
Rank    Country  Score
1      Finland  7.769
2      Denmark  7.600
3      Norway   7.554
4      Iceland  7.494
5  Netherlands  7.488

=====
BOTTOM 5 LEAST HAPPY COUNTRIES (2019)
=====
Rank    Country  Score
156  South Sudan  2.853
155  Central African Republic  3.083
154    Afghanistan  3.203
153      Tanzania  3.231
152        Rwanda  3.334

```

OBJECTIVE 2: CORRELATION ANALYSIS

```
# Calculate correlation with Happiness Score
```

```

numeric_df = df.select_dtypes(include=[np.number])

correlations = numeric_df.corr()['Score'].sort_values(ascending=False)

print("=" * 60)

print("CORRELATION WITH HAPPINESS SCORE (Strongest to Weakest)")

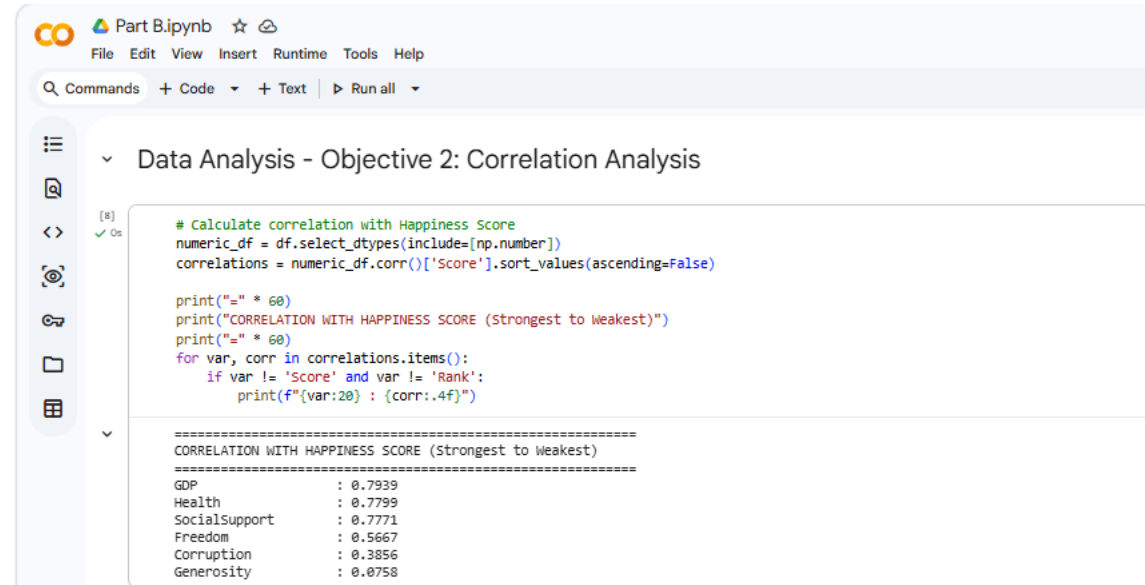
print("=" * 60)

for var, corr in correlations.items():

    if var != 'Score' and var != 'Rank':

        print(f"{var:20} : {corr:.4f}")

```



```

# Calculate correlation with Happiness Score
numeric_df = df.select_dtypes(include=[np.number])
correlations = numeric_df.corr()['Score'].sort_values(ascending=False)

print("=" * 60)
print("CORRELATION WITH HAPPINESS SCORE (Strongest to Weakest)")
print("=" * 60)
for var, corr in correlations.items():
    if var != 'Score' and var != 'Rank':
        print(f"{var:20} : {corr:.4f}")

```

```

=====
CORRELATION WITH HAPPINESS SCORE (Strongest to Weakest)
=====
GDP                        : 0.7939
Health                    : 0.7799
Socialsupport              : 0.7771
Freedom                   : 0.5667
Corruption                 : 0.3856
Generosity                 : 0.0758

```

Key Findings:

- **GDP (Economy)** has the strongest correlation with happiness ($r = 0.7939$)
- **Health** follows closely ($r = 0.7799$)
- **Social Support** is also very strong ($r = 0.7771$)
- **Generosity** shows almost no correlation ($r = 0.0758$)

OBJECTIVE 3: REGIONAL COMPARISON

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```
# Create region categories based on geography
def assign_region(country):
    europe = ['Finland', 'Denmark', 'Norway', 'Iceland', 'Netherlands', 'Switzerland',
              'Sweden', 'Austria', 'Belgium', 'Germany', 'France', 'Spain', 'Italy',
              'United Kingdom', 'Ireland', 'Luxembourg', 'Malta', 'Czech Republic',
              'Slovakia', 'Poland', 'Hungary', 'Romania', 'Bulgaria', 'Greece',
              'Portugal', 'Croatia', 'Serbia', 'Bosnia', 'Albania', 'North Macedonia',
              'Montenegro', 'Slovenia', 'Estonia', 'Latvia', 'Lithuania', 'Ukraine',
              'Belarus', 'Moldova', 'Russia', 'Cyprus']

    asia = ['China', 'India', 'Japan', 'South Korea', 'Indonesia', 'Pakistan',
            'Bangladesh', 'Philippines', 'Vietnam', 'Thailand', 'Malaysia', 'Nepal',
            'Sri Lanka', 'Myanmar', 'Mongolia', 'Kazakhstan', 'Uzbekistan', 'Tajikistan',
            'Kyrgyzstan', 'Turkmenistan', 'Laos', 'Cambodia', 'Afghanistan', 'Iran',
            'Iraq', 'Saudi Arabia', 'United Arab Emirates', 'Qatar', 'Kuwait', 'Bahrain',
            'Jordan', 'Lebanon', 'Israel', 'Palestinian Territories', 'Syria', 'Yemen']

    africa = ['South Africa', 'Nigeria', 'Egypt', 'Ethiopia', 'Kenya', 'Tanzania',
              'Uganda', 'Algeria', 'Morocco', 'Sudan', 'Ghana', 'Angola', 'Mozambique',
              'Madagascar', 'Cameroon', 'Niger', 'Burkina Faso', 'Mali', 'Malawi',
              'Zambia', 'Somalia', 'Chad', 'South Sudan', 'Rwanda', 'Burundi',
              'Benin', 'Tunisia', 'Libya', 'Liberia', 'Mauritania', 'Botswana',
              'Namibia', 'Gabon', 'Lesotho', 'Zimbabwe', 'Congo', 'Swaziland',
              'Mauritius', 'Sierra Leone', 'Togo', 'Comoros', 'Guinea', 'Gambia',
              'Senegal', 'Ivory Coast']

    americas = ['United States', 'Canada', 'Mexico', 'Brazil', 'Argentina', 'Chile',
                'Colombia', 'Peru', 'Venezuela', 'Ecuador', 'Bolivia', 'Paraguay',
                'Uruguay', 'Guyana', 'Suriname', 'French Guiana', 'Costa Rica',
                'Guatemala', 'Honduras', 'Nicaragua', 'El Salvador', 'Panama',
                'Cuba', 'Haiti', 'Dominican Republic', 'Jamaica', 'Trinidad']

    oceania = ['Australia', 'New Zealand']

    if country in europe: return 'Europe'
    elif country in asia: return 'Asia'
    elif country in africa: return 'Africa'
    elif country in americas: return 'Americas'
    elif country in oceania: return 'Oceania'
    else: return 'Other'

df['Region'] = df['Country'].apply(assign_region)

# Calculate regional averages
regional_avg = df.groupby('Region')['Score'].agg(['mean', 'count']).sort_values('mean', ascending=False)
print("=" * 60)
print("REGIONAL AVERAGE HAPPINESS SCORES")
print("=" * 60)
print(regional_avg.round(3))
```

Output:

Part B.ipynb

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```
=====
REGIONAL AVERAGE HAPPINESS SCORES
=====
              mean  count
Region
Oceania    7.268      2
Europe     6.295     39
Americas   6.046     22
Other       5.239     15
Asia        5.211     36
Africa      4.387     42
```

DATA ANALYSIS – VISUALIZATION	
Code	Output

```
# Set style for better visuals
```

```
plt.style.use('seaborn-v0_8-darkgrid')
```

```
sns.set_palette("viridis")
```

```
# FIGURE 1: Top 10 Happiest Countries
```

```
fig, axes = plt.subplots(2, 2, figsize=(14, 12))
```

```
# Plot 1: Top 10 Happiest Countries
```

```
top10 = df.nlargest(10, 'Score')
```

```
axes[0, 0].barh(top10['Country'], top10['Score'],
color='green')
```

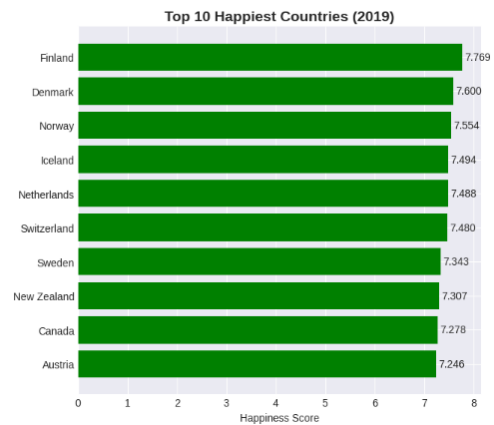
```
axes[0, 0].set_xlabel('Happiness Score')
```

```
axes[0, 0].set_title('Top 10 Happiest Countries (2019)',
fontsize=14, fontweight='bold')
```

```
axes[0, 0].invert_yaxis()
```

```
for i, v in enumerate(top10['Score']):
```

```
    axes[0, 0].text(v + 0.05, i, f'{v:.3f}', va='center')
```



```
# Plot 2: Bottom 10 Least Happy Countries
```

```
bottom10 = df.nsmallest(10, 'Score')
```

```
axes[0, 1].barh(bottom10['Country'], bottom10['Score'],
color='red')
```

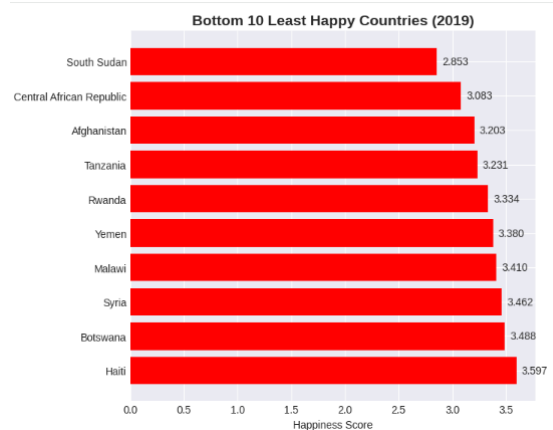
```
axes[0, 1].set_xlabel('Happiness Score')
```

```
axes[0, 1].set_title('Bottom 10 Least Happy Countries
(2019)', fontsize=14, fontweight='bold')
```

```
axes[0, 1].invert_yaxis()
```

```
for i, v in enumerate(bottom10['Score']):
```

```
    axes[0, 1].text(v + 0.05, i, f'{v:.3f}', va='center')
```

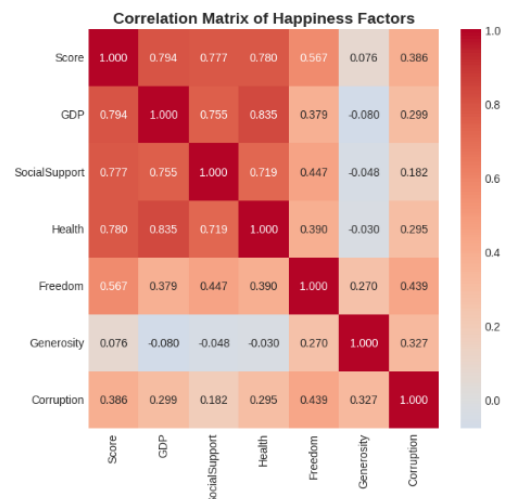


```
# Plot 3: Correlation Heatmap
```

```
corr_matrix = df[['Score', 'GDP', 'SocialSupport', 'Health',
'Freedom', 'Generosity', 'Corruption']].corr()
```

```
sns.heatmap(corr_matrix, annot=True,
cmap='coolwarm', center=0, ax=axes[1, 0], fmt='.3f')
```

```
axes[1, 0].set_title('Correlation Matrix of Happiness
Factors', fontsize=14, fontweight='bold')
```



Plot 4: Factor Comparison - Top 5 vs Bottom 5

```
factors = ['GDP', 'SocialSupport', 'Health', 'Freedom']

top5_avg = df.nlargest(5, 'Score')[factors].mean()

bottom5_avg = df.nsmallest(5, 'Score')[factors].mean()

x = range(len(factors))

width = 0.35

axes[1, 1].bar([i - width/2 for i in x], top5_avg, width,
label='Top 5 Happiest', color='green', alpha=0.7)

axes[1, 1].bar([i + width/2 for i in x], bottom5_avg, width,
label='Bottom 5 Least Happy', color='red', alpha=0.7)

axes[1, 1].set_xticks(x)

axes[1, 1].set_xticklabels(factors)

axes[1, 1].set_ylabel('Average Score')

axes[1, 1].set_title('Factor Comparison: Top 5 vs Bottom
5 Countries', fontsize=12, fontweight='bold')

axes[1, 1].legend()

plt.tight_layout()

plt.savefig('happiness_analysis_figures.png', dpi=300,
bbox_inches='tight')

plt.show()
```

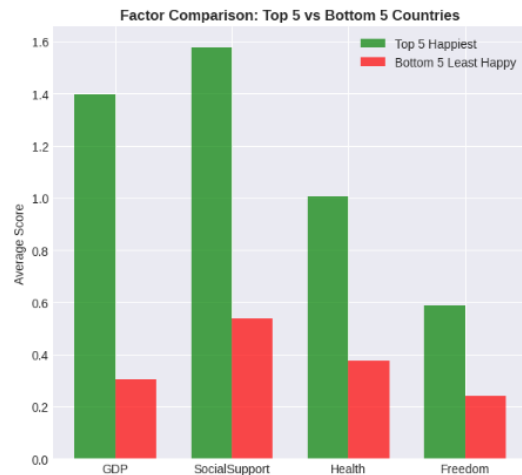


FIGURE 5: Scatter Plot - GDP vs Happiness

```
plt.figure(figsize=(10, 6))

scatter = plt.scatter(df['GDP'], df['Score'], c=df['Score'],
cmap='viridis', s=100, alpha=0.7)

plt.colorbar(scatter, label='Happiness Score')

plt.xlabel('GDP per Capita (log scale)', fontsize=12)

plt.ylabel('Happiness Score', fontsize=12)

plt.title('GDP vs Happiness Score (2019)', fontsize=14,
fontweight='bold')

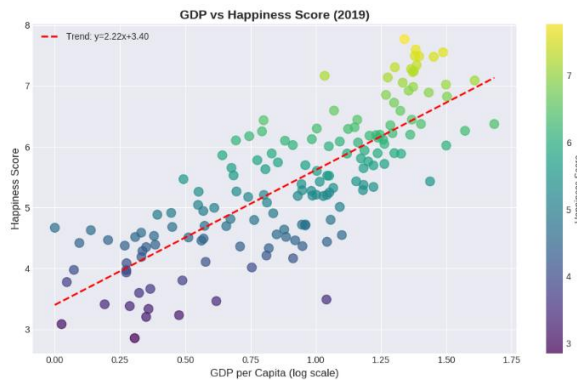
# Add trend line

z = np.polyfit(df['GDP'], df['Score'], 1)

p = np.poly1d(z)

plt.plot(df['GDP'].sort_values(),
p(df['GDP'].sort_values()), "r--", linewidth=2,
label=f'Trend: y={z[0]:.2f}x+{z[1]:.2f}')

plt.legend()
```




```
plt.tight_layout()

plt.savefig('gdp_happiness_scatter.png', dpi=300)

plt.show()
```

EXECUTIVE SUMMARY

Based on the analysis of 156 countries from the World Happiness Report 2019, the following key findings emerge:

Key Findings:

1. Nordic countries dominate happiness rankings

- Finland (7.769), Denmark (7.600), Norway (7.554), Iceland (7.494), and Netherlands (7.488) are the top 5 happiest nations
- These countries share common characteristics: strong social safety nets, low corruption, high GDP, and excellent healthcare

2. Economic prosperity is the strongest predictor of happiness

- GDP per capita shows a very strong correlation ($r = 0.7939$) with happiness scores
- The top 5 happiest countries have significantly higher GDP than bottom 5 countries

3. Social support and health are equally critical

- Social support ($r = 0.7771$) and healthy life expectancy ($r = 0.7799$) are nearly as important as GDP
- These three factors form the "happiness trifecta"

4. Generosity has minimal impact on national happiness

- Correlation of only 0.0758 suggests that while individual giving is virtuous, it doesn't drive national happiness scores

5. Significant regional disparities exist

- Oceania (7.27) and Europe (6.30) far outperform Africa (4.39) and Asia (5.21)
- Conflict, poverty, and lack of infrastructure explain most of the gap

6. Corruption perception matters

- Correlation of 0.3856 indicates countries with lower corruption perception have higher happiness

Limitations of This Analysis:

1. **Correlation $\neq$ Causation:** High GDP correlates with happiness but doesn't necessarily cause it

2. **Self-reported data:** Happiness scores are subjective and culturally biased
3. **Single year (2019):** Pre-COVID snapshot; current rankings may have changed
4. **Regional categorization:** Some countries may fit multiple regions

REFERENCES

1. World Happiness Report 2019. (2019). *World Happiness Report*. Sustainable Development Solutions Network. <https://worldhappiness.report/ed/2019/>
2. Helliwell, J., Layard, R., & Sachs, J. (2019). *World Happiness Report 2019*, New York: Sustainable Development Solutions Network. <https://www.worldhappiness.report/ed/2019/>
3. Dataset: World Happiness Report 2019 CSV. Retrieved from Kaggle: <https://www.kaggle.com/datasets/unsdsn/world-happiness>